

KGOPOTSO POOPEDI

Software Development | Mobile Development | Database Development and Administration

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PROFESSIONAL PROFILE

Final-year Diploma in Computer Science student at Tshwane University of Technology seeking a Work Integrated Learning (WIL) opportunity. Experienced in designing and developing desktop, web, and IoTbased software solutions using Java, C#, JavaScript, HTML, CSS, SQL, and MySQL. Hands-on experience building multi-tier applications, designing relational databases, integrating Arduino sensors, and implementing AI-assisted features for real-world projects. A motivated problem solver who enjoys learning new technologies and working collaboratively to deliver practical software solutions.

CORE COMPETENCIES

Software Development	Web & Mobile App Development	Full-Stack Web Development
Object-Oriented Programming	Database Design & SQL	Multi-Tier Architecture
Excel, Reporting & Google Workspace	Time Management	Attention to Detail & Adaptability

My Portfolio : <https://debug-kgopotso.github.io/Portfolio/>

KEY PROJECTS

Water Monitoring System | Full-Stack Developer Team Project | 2026 ○ Designed and implemented a MySQL relational database to store water quality measurements, tank levels, user information, and historical sensor data.

- Developed an Arduino-based IoT solution that captured real-time readings from a TDS (Total Dissolved Solids) sensor to monitor water quality and an ultrasonic sensor to measure the water level inside a JoJo tank.
- Built a multi-tier web application that displayed live sensor readings, historical data, and water quality information through an interactive web interface.
- Implemented an AI prediction feature that estimated the remaining time before the JoJo tank would run out of water based on historical water usage and current tank levels.
- Worked collaboratively within a development team using Git and GitHub for version control and project management.

Technologies: JavaScript, HTML, CSS, MySQL, Arduino, Git, GitHub **Repository:**

<https://github.com/debug-Kgopotso/WaterMonitorWebAppV1>

Smart City Project | Full-Stack Developer *Solo Project | 2026* ○ Designed and implemented a MySQL relational database to store incident reports, user information, citizen feedback, and system audit logs.

- Developed a Java EE web application that allowed citizens to report city issues including water leaks, potholes, electricity outages, illegal dumping, and broken traffic lights.
- Integrated GPS location services enabling automatic address detection and real-time incident mapping using geolocation APIs.
- Implemented image upload functionality allowing citizens to attach photo evidence to incident reports.
- Built role-based dashboards for Citizens (report incidents, track status), Municipality Workers (view assigned jobs, update progress), and Administrators (heat maps, statistics, response time reports).
- Worked collaboratively within a development team using Git and GitHub for version control and project management.

Technologies: Java EE, JPA, GlassFish, Derby Database, JavaScript, HTML, CSS, Git, GitHub **Repository:** <https://github.com/debug-Kgopotso/Smart-City-Project>

Cyber Security Monitor Project | Full-Stack Developer *Solo Project | 2026* ○ Designed and implemented a Derby relational database to store user information, security events, threat intelligence data, and system audit logs.

- Developed a Java EE web application that provided real-time security monitoring and threat detection capabilities for a Security Operations Center (SOC).
- Built a comprehensive dashboard displaying live security events, threat level statistics (Critical, High, Medium, Low), and failed login attempt tracking.
- Implemented threat intelligence features allowing administrators to block/unblock suspicious IP addresses and track attack patterns including Malware, DDoS, Brute Force, and Phishing.
- Developed an attack simulator that generates fake cyber attacks for demonstration and testing purposes, automatically populating the dashboard with simulated threats. ○ Created role-based access control with three user levels: Admin (full system access), Analyst (threat monitoring and attack generation), and Viewer (read-only personal dashboard).
- Implemented audit logging to track all user activities, failed login attempts, and system changes with IP address and user agent capture.
- Built user management with password strength validation and automatic account locking after 5 failed login attempts.

Technologies: Java EE, GlassFish, Derby Database, HTML, CSS, Git, GitHub **Repository:** <https://github.com/debug-Kgopotso/Cyber-Security-MonitoringApp>

EDUCATION

Diploma in Computer Sciences

Expected Completion 2026

Tshwane University of Technology

Key modules: *Software Development | Mobile Development | Database Development and Administration*

National Senior Certificate (Matric)

Completed 2021

Lyttelton Manor High School

LEADERSHIP EXPERIENCE

Headboy (2016)

RCL Committee Member(2021)